Hydration and the Importance of Water

Introduction to Hydration

Hydration means having enough water in the body to carry out its normal functions. Water is the most important nutrient: a person can live for weeks without food but only a few days without water. Water is the main component of the body, and it is involved in almost every process of life. Staying well hydrated improves energy, concentration, digestion, skin health and physical performance, while even mild dehydration can cause tiredness, headaches and poor focus. In hot, humid countries like India, and especially in cities such as Chennai, where temperatures are high for much of the year, proper hydration is particularly important for students, workers and athletes.

What Percentage of the Body Is Water?

Water makes up roughly 60 percent of the body weight of an average adult. The percentage varies between about 50 and 65 percent depending on age, sex and body composition. Newborn babies are about 75 percent water, and older people have a lower percentage. Men generally have a higher water percentage than women because muscle holds more water than fat. Different organs contain different amounts: the brain and heart are about 73 percent water, the lungs about 83 percent, the skin about 64 percent, the muscles and kidneys about 79 percent, and even the bones are about 31 percent water. Blood plasma is about 90 percent water.

Functions of Water in the Body

Water regulates body temperature through sweating, since evaporation of sweat cools the body. It carries nutrients and oxygen to cells through the blood, helps digest food and absorb nutrients, removes waste products through urine and sweat by way of the kidneys, lubricates joints, cushions the brain and spinal cord, keeps the skin healthy, moistens the eyes, nose and mouth, and maintains blood pressure and blood volume. Chemical reactions inside cells need water. The kidneys filter around 180 litres of blood plasma a day to produce about 1 to 2 litres of urine.

How Much Water Should an Adult Drink Daily?

There is no single number that suits everyone, since requirements depend on body size, climate, activity, diet and health. A common guideline for adults is to drink about 2 to 3 litres of fluids a day, which is about 8 to 12 glasses. The US National Academies suggest a total water intake, from drinks and food, of about 3.7 litres for men and 2.7 litres for women daily, since about 20 percent comes from food such as fruits, vegetables and curd rice. In hot weather, in a tropical climate like Tamil Nadu, or during exercise, one needs more. Pregnant and breastfeeding women need extra. A simple practical guide is to drink enough so that urine is pale yellow and to drink regularly rather than waiting for thirst. Water, buttermilk, coconut water, lemon water, and fruit-rich foods all count, while tea and coffee count in moderation. Sugary soft drinks and energy drinks are not good choices.

What Are the Signs of Dehydration?

Dehydration occurs when the body loses more water than it takes in, through sweating, vomiting, diarrhoea, fever or not drinking enough. Mild to moderate signs include thirst, dry mouth and lips, dark yellow urine and less frequent urination, tiredness, headache, dizziness or lightheadedness, dry skin, muscle cramps and difficulty concentrating. Severe dehydration is a medical emergency and signs include very little or no urination, sunken eyes, rapid heartbeat and breathing, confusion, extreme weakness, cold hands and feet and fainting. Infants and elderly people are particularly at risk. The colour of urine is a quick indicator: pale straw colour is healthy, while dark amber suggests dehydration. Heat exhaustion and heat stroke can follow dehydration in summer.

Oral Rehydration Solution (ORS)

In diarrhoea, vomiting or heavy sweating, the body loses salts as well as water, so plain water is not enough. Oral Rehydration Solution (ORS) is a mixture of clean water, glucose and salts such as sodium chloride and potassium chloride, which lets the intestine absorb water efficiently. The Indian doctor Dilip Mahalanabis used it to save thousands of lives in refugee camps during the 1971 Bangladesh war, and the medical journal The Lancet called ORS potentially the most important medical advance of the twentieth century. Packets of ORS are available at pharmacies and government health centres, and a home version can be made with a litre of clean water, six level teaspoons of sugar and half a teaspoon of salt. ORS Week is observed in India in July.

Traditional Hydrating Drinks in Tamil Nadu and India

Tamil Nadu has many traditional hydrating drinks. Neer mor (spiced buttermilk), elaneer (tender coconut water), nannari sarbath, panagam, kanji (rice water), lemon juice and jigarthanda in Madurai help people stay cool in summer. Tender coconut water contains natural electrolytes such as potassium. Fruits such as watermelon, muskmelon, cucumber and oranges have more than 85 to 90 percent water.

Overhydration and Quality of Water

Drinking far too much water in a short time can dilute the sodium in the blood, a condition called hyponatraemia, which is rare but dangerous, mostly in endurance athletes. Water must also be safe. Drinking contaminated water causes diseases like cholera, typhoid and jaundice, so water should be boiled or purified. In Indian homes, steel, glass or clay pots (matkas) are used, and carrying a reusable bottle reduces plastic waste.

Habits for Good Hydration

Start the day with a glass of water, keep a bottle within reach, drink a glass before each meal, set reminders on the phone, drink extra in hot weather and when exercising, eat water-rich foods, and avoid relying only on thirst.

Challenges and Future

Challenges include access to safe drinking water in many parts of India and the habit of drinking sugary beverages. Smart water bottles and awareness campaigns are helping.

Quick Facts

- Water makes up about 60 percent of an adult's body weight.
- Signs of dehydration include thirst, dark urine, dry mouth, headache, tiredness and dizziness.
- Adults commonly need about 2 to 3 litres of fluids a day, more in hot weather.
- ORS replaces water and salts lost in diarrhoea; Dr. Dilip Mahalanabis pioneered its use in 1971.
- Pale yellow urine is a sign of good hydration.